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Standardization can be good for exploration: a social capital view of the productivity dilemma in operational teams

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ABSTRACT

Process Management Programs (PMPs) rely heavily on the standardization of activities to improve organizational efficiency and reliability. Standardization, however, can lead to rigidity and inertia, which may eventually inhibit exploration and innovation. While scholars have provided cognitive and routine-based arguments to cope with this dilemma, the fact that standardization is a social process that requires collaboration between individuals has been overlooked. In this paper, we propose that standardization can improve a team's social capital and thereby facilitate the activities that enable exploration. Specifically, we hypothesize that standardization relates positively to external communication, psychological safety, and the perception of support for innovation in operational teams. These three social capital attributes mediate the relationship between standardization and exploration. We tested this multilevel mediation model by analysing 431 members of 62 operational teams in a large multinational mining company that had recently implemented a Process Management Program (Lean Management). Our multilevel mediation results confirmed that standardisation promoted exploration by enhancing the teams' social capital. For robustness we tested our model with a second survey of 450 workers of different companies and obtained similar results. Therefore, we propose that the productivity dilemma can be balanced by improving operational teams' social capital.

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1. Introduction

Standardization refers to the extent to which work activities and decisions are dictated by rules and procedures (Gilson et al. 2005). While it is widely accepted that standardization positively impacts organizational efficiency and reliability, there is little consensus on how standardization affects an organization's capacity to innovate in the long run (Adler et al. 2009; Backström 2017; Benner and Tushman 2003; Kristensen 2020; Roldán-Bravo et al. 2017; Shalley and Gilson 2017; Sitkin, Sutcliffe, and Schroeder 1994; Zhang, Linderman, and Schroeder 2012). For instance, Motorola and 3M achieved outstanding performance by adopting process management programs (PMPs) that relied on formal procedures and routines. However, as these companies lost their innovative edge, standardization was blamed for diminishing their capacity to explore new opportunities (Canato, Ravasi, and Phillips 2013; Hindo, 2007). In contrast, companies such as Toyota and Pixar rely on standardized procedures and routines to improve both operational efficiency and innovation (Adler et al. 2009; Catmull and Wallace 2014; Wright, Sturdy, and Wylie 2012). These organizational examples provide contradictory evidence regarding the effect of standardization on innovation.

In the literature, there is a long-lasting debate about the effect of standardization on exploration, that is, the organizational capacity for searching for new opportunities. This

debate stems from two different perspectives on standardization: as a *dualism* and as a *duality* (Farjoun 2010). The dualism viewpoint, which comes mainly from the management literature, argues that standardization works well for specialized tasks but constrains discretionary behaviours and cognitive abilities to engage in non-routine tasks required for exploration (Benner and Tushman 2015; Burns and Stalker 1961; Roldán-Bravo et al. 2017; Zhang et al. 2012). In other words, the dualism view maintains that standardisation is detrimental to exploration. The duality perspective, in contrast, acknowledges that while in theory standardisation and exploration may be at odds, empirically they are interdependent and mutually enabling over time (Adler et al. 1999; Farjoun 2010; Kim, Kumar, and Kumar 2012; Kristensen 2020; Smith and Lewis 2011; Zeng et al. 2017).

Duality scholars have used different theoretical lenses to explain the potential positive effect of standardization on companies' or individuals' capacity to explore. For example, its positive effects have been examined from the perspectives of organizational learning (Argote and Miron-Spektor 2011), dynamic capabilities (Anand, Ward, and Tatikonda, 2010; Hansen and Møller 2016), causal ambiguity (Adler et al. 2009; Kristensen 2020; Patel 2011), liabilities of newness (Shashi et al. 2019; Sine, Mitsuhashi, and Kirsch 2006), cumulative capabilities (Ehie and Schoenherr 2020; Sum, Singh, and Heng 2012), project portfolio management (Petro et al.

2020), and organizational routines (Feldman and Rafaeli 2002). Although these studies have brought undeniable benefits to the field of organization studies, they do not account for the social dynamics that enable exploration.

Exploration is a collective process shaped by the quantity and quality of relationships within an organisation (Argote and Miron-Spektor 2011). As such, the literature on the effects of standardization on exploration could benefit from examining the social team dynamics that enable groups to explore.

We build on social capital theory to propose that a team's social capital mediates the relationship between standardization and exploration. Social capital refers to the quality and quantity of interactions between actors in a social network (Kwon and Adler 2014). According to Nahapiet and Ghoshal (1998), social capital has three dimensions: structural, relational, and cognitive. While the structural dimension determines the number of interactions within social networks, the relational and cognitive dimensions qualitatively assess those interactions.

We argue that standardization can improve these three dimensions of team social capital. First, it can enhance the structural dimension by increasing the awareness of 'who knows what' within a network and by encouraging the emergence of shorthand languages that facilitate external communications (Argote and Miron-Spektor 2011). Second, it can improve the relational dimension by positively influencing the shared understanding of workers' responsibilities and entitlements (Bresman and Zellmer-Bruhn 2013). Finally, it can enhance the cognitive dimension by aligning shared perceptions of exploratory behaviours' appropriateness and legitimacy (Kristensen 2020; Wright, Sturdy, and Wylie 2012). We hypothesize that these three social capital dimensions mediate the relationship between standardization and exploration. The structural dimension facilitates access to external knowledge. The cognitive and relational dimensions motivate team members to act upon the exploration opportunities provided by these external sources of knowledge (Nahapiet and Ghoshal, 1998).

To assess the mediation effect of a team's social capital between standardization and exploration, we analysed 431 individuals belonging to 62 operational teams within a large multinational mining company that had recently implemented Lean Management, which fosters highly formalized activities. Specifically, we argue and empirically assess whether standardization can improve the structural, relational, and cognitive dimensions of a team's social capital, which can, in turn, enable and stimulate operational teams to explore. As a robustness check, we also tested our model with a second survey on a more diverse sample of 450 workers working for different companies and obtained similar results.

The results of this study show *how* standardization can improve exploration in operational teams. We find that two novel factors, the cognitive and structural dimensions of social capital, mediate the relationship between standardization and exploration. By examining these relationships, we put forward an original argument to explain how both ideas

and actions can be concurrently elicited in teams. We thus respond to calls to 'question ... how to balance the need for creativity and standardization' (Shalley and Gilson 2017, p.10), and discuss further implications for advancing theory and practice.

The paper is structured as follows. In Section 2, we review the literature on standardization and exploration. We then propose a model whereby a team's social capital mediates the relationship between standardisation and exploration (Section 3). Our data, methods and results are discussed in Section 4. Section 5 outlines the results of our model, and lastly, we discuss the implications, limitations, and future avenues of research (Section 6).

2. Theoretical framework

The standardization of activities has long been fundamental to operation management, as it promotes exploitation, that is, the organizational capacity for achieving higher efficiency and reliability (Adler et al. 2009; Benner and Tushman 2015). Standardization acts as a coordination mechanism that allows the control of unwanted variances (Dobrzykowski, McFadden and Vonderembse 2016), the refinement of existing competencies (Argote and Miron-Spektor 2011), the interchangeability of employees (Briscoe 2007), reduction in defects (Panwar et al. 2015), and the scalability of activities (Gittell and Douglass 2012), all of which promote exploitation. Therefore, diverse PMPs, including Lean, Total Quality Management (TQM), Six Sigma, and Reengineering, have adopted standardization as a core mechanism for improving organizational efficiency and reliability. The influence of standardization on exploration and innovation, however, remains unclear.

2.1. Standardization and exploration as a dualism

Its positive effects notwithstanding, the standardization of activities can constrain the explorative capabilities required for change and innovation, creating what has been called 'the productivity dilemma' (Abernathy 1978; Backström 2017; Benner and Tushman 2003; Roldán-Bravo 2017; Zhang et al. 2012). This dilemma has emerged from scholars' description of exploitative and explorative activities as a dualism. In other words, although these activities complement each other, they are nevertheless incompatible and mutually exclusive (Farjoun 2010; Smith and Lewis, 2011). Exploitation activities—such as those enabled by standardization—are associated with efficiency, reliability, and productivity; thus, they require routine, discipline, and control. In contrast, exploration activities are associated with searching for, evaluating, and experimenting with new opportunities; thus, they require openness, flexibility, and discretion (March 1991). Therefore, the mindset, attitudes, and processes needed to exploit and explore are inconsistent with each other (Abernathy 1978; Benner and Tushman 2015).

Several theoretical explanations have been proposed as to why standardization can hinder exploration. Based on Max Weber's *iron cage*, initial studies posited that bureaucratic

elements achieve organizational efficiency and reliability at the price of individuals' alienation (Burns and Stalker 1961). More recent studies have argued that standardisation could restrain discretionary behaviours that motivate non-routine activities (Adler 2012) and restrict information search (Benner and Tushman 2015). Other studies indicate that standardisation can reduce the need for individuals' interactions by codifying tacit knowledge (Mom, Van den Bosch, and Volberda 2009) and narrow individuals' perspectives by instilling concepts of appropriate behaviours (Hirst et al. 2011). Standardization can also be expected to foster expertise in delimited areas (Huo, Han, and Prajogo 2014), motivate short-term efficiency prioritisation (Terziovski and Guerrero 2014) and lead to a mechanical execution of activities impairing individuals' ability to think what could be done differently (Benner and Tushman 2015). Therefore, these effects of standardisation can constrain opportunities, motivation, and capabilities to explore (Farjoun 2010).

Nevertheless, empirical evidence has been inconclusive regarding the relationship between standardisation and exploration, delivering inconsistent and sometimes paradoxical findings. For instance, several studies failed to find a negative relationship between standardization and explorative outcomes at the organizational level (Jansen, Van den Bosch, and Volberda 2005; Ruiz-Moreno, Tamayo-Torres, and García-Morales 2015). At an individual level, a range of studies finds no evidence to support the proposition that standardisation inhibits innovative behaviours such as individual ambidexterity (Mom, Van den Bosch, and Volberda 2009), individual initiatives to make changes (Raub 2008), or individual creativity (Hirst et al. 2011). In contrast, evidence has indicated a positive relationship between standardization and the innovative performance of firms (Chen, Damanpour, and Reilly 2010; Damanpour, 1991; Evanschitzky et al. 2012; Kim, Kumar, and Kumar 2012; Kuratko, Hornsby, and Covin 2014; Ng et al. 2015; Storey et al. 2016; Zeng et al. 2017), as well as between standardisation and the explorative behaviours of individuals (Briscoe 2007; Escrig-Tena et al. 2018; Kristensen 2020; Yang, Lee, and Cheng 2017).

2.2. Standardization and exploration as a duality

In the light of the discrepancies between theoretical frameworks and empirical results, another stream of research began to view the relationship between standardization and exploration as a duality (Adler et al. 2009; Kim, Kumar, and Kumar 2012; Kristensen 2020; Patel 2011; Ramus, Vaccaro, and Brusoni 2017). A duality approach describes contradictory elements and tensions within an object of study, but it also recognises them as interdependent and mutually enabling rather than separated and opposed to each other (Farjoun 2010). Contrary to the dualism standpoint, the duality perspective does not assume a unique relationship between mechanisms and outcomes, such as standardisation and exploitation; instead, it proposes that these mechanisms may also produce outcomes contradictory to their nature (Farjoun 2010; Smith and Lewis 2011). In other words, the duality perspective posits that standardisation not only

promotes exploitation but can also enable exploration (Smith and Lewis 2011).

Scholars adopting this duality perspective have provided cognitive and routine-based arguments on how standardization enables individuals to explore. Standardization can increase creativity by reducing the need for trial and error, thus freeing up resources for non-routine activities (Adler et al. 2009). For instance, athletes, musicians, expert chefs, and artists follow disciplined regimes to stimulate creativity and achieve performance excellence (Winter in Adler et al. 2009; Albors-Garrigós, Martínez-Monzo, and García-Segovia 2018). Similarly, scientists use formalized procedures to produce robust theories by minimizing errors and unwanted deviations. Standardization also requires workers to discuss 'what,' 'how,' and 'why' work is done (Anand, Ward, and Tatikonda 2010; Linderman et al. 2004). These discussions enable the transformation of implicit into explicit knowledge, broadening workers' knowledge base (Anand, Ward, and Tatikonda 2010; Boon Sin et al. 2015; Cho and Linderman 2019) and reducing causal ambiguity in organizational processes (Patel 2011). A broad knowledge base, in turn, favours the combination of existing knowledge and increases the capacity to acquire and synthesise new knowledge (Cohen and Levinthal 1990). A greater depth of knowledge about one's work can also increase self-competence (De Treville and Antonakis 2006) and motivate workers to engage in non-routine activities (Yang, Lee, and Cheng 2017). Moreover, the standardization of non-routine activities, such as problem solving and idea generation, can offer significant benefits. Standardization provides a general-purpose meta-routine that helps individuals in the adaptation of existing activities (Adler et al. 1999; Escrig-Tena et al. 2018), thereby increasing organizational dynamic capabilities (Anand, Ward, and Tatikonda 2010).

However, cognitive or routine-based explanations of the effect of standardization on exploration fail to recognize that innovation is a collective endeavour. An organization's capacity to explore does not reside in any one individual but depends on the coordination capacity across a spectrum of individual capabilities (Argote and Miron-Spektor 2011; Cohen and Levinthal, 1990). For exploration to occur, individuals must have access to information from others (Argote and Miron-Spektor 2011) and be immersed in interpersonal environments that motivate and mobilize explorative actions (Eschenbächer et al. 2011; Kwon and Adler 2014). Therefore, to assess the effects of standardization on exploration, we need to focus on social interactions, knowledge-sharing networks, norms, and culture that mobilize and motivate people to explore. In other words, on how standardization affects the social capital of operational teams.

2.3. Social capital

Social capital theory provides a perspective to examine social relationships and how these confer advantages to groups (Carey, Lawson, and Krause 2011). Put simply, social capital theory represents the idea that social networks can affect performance (Arrow 2000). Specifically, social capital

describes ‘assets and resources made available through relationships within the social structure and content of an actor’s social relationships’ (Payne et al. 2011, p. 497).

Initial studies on social capital focussed on individual behaviours, arguing that the extent to which actors’ contacts are themselves connected facilitates the emergence of trust, thereby providing access to assets and shared resources (Coleman 1988). Subsequent studies expanded this view to account for those relationships’ network and structure, which offer knowledge brokerage opportunities (Burt 2001). Thus, the value of the resources derived from connections spans immediate contacts (Han et al. 2018), and sparse networks with few redundant ties often provide greater social capital benefits (Burt, 2001).

Individual perspectives on social capital led way to teams (Oh et al. 2006), business units (Tsai and Ghoshal 1998), organisations (Inkpen and Tsang, 2005), and communities (Putnam 2005). While situated at various levels of analysis, most (if not all) views on social capital agree that social relationships are important for performance outcomes but can potentially have adverse effects (Seragaldin and Grootaert 2000). Thus, social capital effects vary depending on the unit of analysis and how these units are connected and nested within particular social contexts.

Social capital is, therefore, not a monolithic concept (cf. Putnam 2005), and several distinctions are made in the literature. Granovetter (1992) distinguishes between a network’s configuration, *structural embeddedness*, and the quality of the relationships in this network, *relational embeddedness*. In the framework of Nahapiet and Ghoshal (1998), these forms of embeddedness can be categorised along three dimensions of social capital: structural, relational, and cognitive.

The *structural dimension* refers to the configuration of linkages between entities, based on whom entities reach and how entities reach them (Nahapiet and Ghoshal, 1998). Team members’ social ties outside their group affect the opportunities to exchange information and support the team’s effectiveness (Oh, Labianca, and Chung 2006). This structural dimension can be operationalized by assessing team members’ frequency of communications with other groups (Atuahene-Gima and Murray 2007; Villena et al. 2011). Groups that communicate more frequently with people in other units have greater access to resources and information (Alletto et al., 2017), which allows capturing the opportunity of individuals within groups to explore (Oh, Labianca, and Chung 2006).

The *relational dimension* refers to ‘the trust, obligation, and identification present in personal relationships between people’ (Carey, Lawson, and Krause, 2011, p. 278). In teams, the relational dimension depends on the individual relationships within the group (Oh, Labianca, and Chung 2006). Ingroup relationships can be characterized by psychological safety, that is, the ‘perceptions of risk or vulnerability, as well as making choices to minimize negative consequences, and both have potential positive consequences for work groups.’ (Edmondson 2004). The main difference between trust and psychological safety is that the former pertains primarily to a dyadic relationship –between individuals or collectives such

as firms; the latter being a team-level construct (Newman, Donohue, and Eva 2017). Therefore, psychological safety can represent a dimension of social capital within the team context and allows capturing the motivation of individuals within groups to explore (Lee, Swink, and Pandejpong 2011). The interplay between relational and structural forms of social capital is positively linked with managerial performance, including innovation outcomes (Moran 2005).

For the *cognitive dimension*, we draw on social cognitive theory insights to explain how individuals acquire and maintain behaviours, considering the social environment in which those behaviours are performed (Inkpen and Tsang, 2005). The cognitive dimension symbolises how workers make sense of organisational information (Lechner, Frankenberger, and Floyd 2010). It is represented by languages, codes, and narratives (Lee 2009) that reflect shared goals, vision, and values between actors embedded in a social system (Tsai and Ghoshal 1998). This cognitive dimension has been operationalized along a wide range of constructs, including whether individuals hold a shared vision or follow prevailing work paradigms (Lechner, Frankenberger, and Floyd 2010; Tsai and Ghoshal 1998), the shared value system among operators (Handoko, Bresnen, and Nugroho 2018), or the common languages used among workers (Chang and Chuang, 2011). It has also been operationalized as the tenure of individuals within an organization (Polyviou, Croxton, and Knemeyer 2019), the understanding of strategic goals among workers (Atuahene-Gima and Murray 2007), and the possession of a shared culture (Villena, Revilla, and Choi 2011). For teams to explore, they need to perceive that the organization expects and approves workers to do things differently in the work environment (West 1990).

For exploration activities to be perceived by the group as legitimate, these should reflect the organisation’s rules, norms, or culture (Inkpen and Tsang, 2005). As argued by James et al. (1977) and Scott and Bruce (1994), the organizational climate sets expectations for the accepted range of behaviours within an organization. When the organizational climate is supportive of innovation, innovativeness will be perceived as culturally appropriate, in turn legitimizing experimentation (Yuan and Woodman 2010). Higher perceived support for innovation will also result in higher levels of trust among team members (Zhang et al. 2007). Support for innovation is thus an appropriate proxy for cognitive social capital, representing communication between actors sharing codes and narratives (Lee and Jones 2008; Nahapiet and Ghoshal 1998). This construct has been widely examined as a driver of explorative outcomes in team literature (Hülsheger, Anderson, and Salgado 2009). Thus, it represents a relevant construct to assess the ‘shared interpretations and systems of meaning’ related to exploration among team members (Nahapiet and Ghoshal 1998).

3. Theoretical model

Building on the duality perspective and social capital theory, we propose that standardisation can improve the three dimensions of team social capital, which, in turn, can

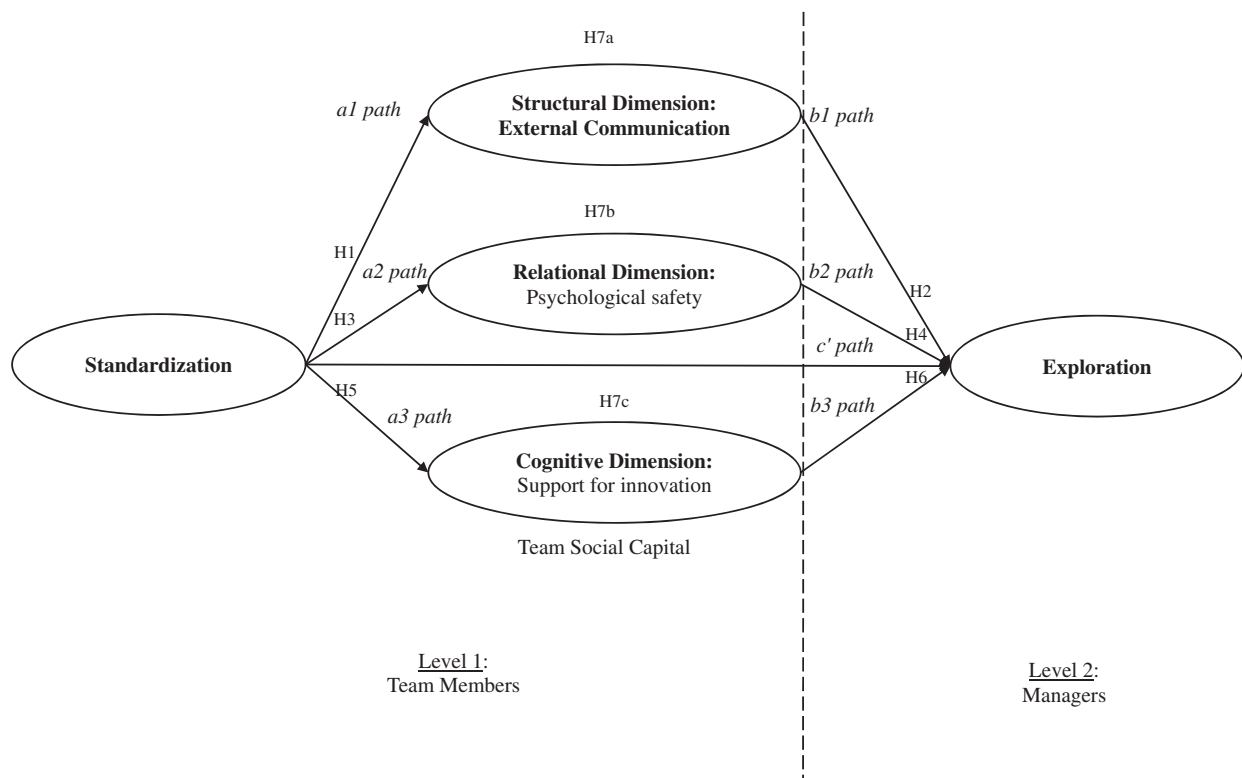


Figure 1. Theoretical multilevel mediation model of the standardisation to exploration.

facilitate collective exploration. Standardisation can increase the frequency of teams' external communications by developing shorthand languages and the awareness of who knows what within organisations (Argote and Ren 2012). Furthermore, Bresman and Zellmer-Bruhn (2013) and Bunderson and Boumgarden (2010) showed that standardization can enhance learning behaviours by improving the quality of team relationships (i.e. psychological safety) within teams. Finally, standardization facilitates aligning shared perceptions of the appropriateness and legitimacy of exploratory behaviours (i.e. support for innovation). Standardization serves as a coordination mechanism of collective work (Adler 2012). It therefore should increase the three dimensions of team social capital, which in turn affects the motivation and opportunities of teams to explore. In the light of the above, we propose a theoretical model whereby a team's social capital mediates the relationship between standardization and exploration (Figure 1). In the following, we elaborate on each path of the proposed mediation model.

3.1. Standardization, team external communications, and exploration

Standardization processes require diverse teams to collectively discuss and codify their tasks, transforming implicit into explicit knowledge (Anand, Ward, and Tatikonda 2010; Cho and Linderman 2019). This collective discussion facilitates the communication between otherwise disconnected teams, and the codification process improves workers' awareness about 'who knows what' within the organization (Argote and Miron-Spektor 2011; Argote and Ren 2012). Greater awareness about whom to consult for particular matters means

that individuals spend less time seeking information (Argote and Ren 2012). Furthermore, the process of discussing tasks reveals their collective nature and their similarities, leading to the creation of 'shorthand languages' that facilitate the transfer of knowledge between different teams (Adler 2012). These discussions also make visible the organizational architecture of that process, facilitating the coordination of activities (Kristensen 2020). Thus, higher awareness and shorthand languages created through the standardization process facilitate the search and exchange of information with external teams (Argote and Miron-Spektor 2011). Hence, we propose the following:

Hypothesis 1: Standardization is positively related to the frequency of external communications in operational teams.

Standardization can lead to team exploration by affecting the search and exchange of information with external teams. If team members are not exposed to external teams, they tend to solve current problems by resorting to what has worked in the past (Adler et al. 2009). Although prior experience can help teams solve problems more efficiently, information redundancy produces local searchers with limited potential for solutions (Cho and Linderman 2019). Conversely, teams with a higher frequency of external communications have better chances of exchanging common problems and best practices (Ancona and Caldwell 1992; Andersen, Ellegaard, and Kragh 2020; Easton and Rosenzweig 2015; Kulangara, Jackson, and Prater 2016). This exchange of information intensifies the divergence of perspectives to understand problems and expands the repertoire of inputs for potential solutions (Chung and Jackson 2013; Maurer, Bartsch, and Ebers 2011). Thus, more externally

communicative teams have a higher potential to explore since ‘whom you know’ affects ‘what you know’ (Kwon and Adler 2014). We therefore hypothesize:

Hypothesis 2: External communication increases exploration in operational teams.

3.2. Standardization, psychological safety, and exploration

Standardization can reduce uncertainty with respect to the relationships within a team (Bresman and Zellmer-Bruhn 2013). The standardization of activities makes it clear who is accountable for each task (Kristensen 2020). An explicit definition of individuals’ responsibilities reduces errors of judgement and biases (Bunderson and Boumgarden 2010; De Clercq, Dimov, and Thongpapanl 2013; Lee, Mazmanian, and Perlow 2020). Assigning responsibilities also reduces the chances of rewards (or punishments) being meted out based on favouritism, promoting a sense of fairness within the team (Gittell and Douglass 2012; Son et al. 2019). In other words, the standardization of activities can increase team members’ confidence that they will receive the same treatment as others in similar situations (Kristensen 2020). Higher certainty about roles, rewards, and responsibilities, in turn, diminishes individual perceptions of risks related to social interactions (i.e., increases psychological safety). Thus, we propose:

Hypothesis 3: Standardization is positively related to psychological safety in operational teams.

We suggest that a mechanism by which standardization leads to team exploration involves psychological safety (Edmondson 1999). To explain this mechanism, we posit that standardization provides certainty about role objectives. Such certainty offers expectable behaviours and fosters trust among operators. Conversely, a team that lacks operational standards delivers less predictable behaviours. Therefore, the consequences of interpersonal actions are more uncertain, which can diminish psychological safety (Bresman and Zellmer-Bruhn 2013). The exploration of new opportunities can entail a personal risk because it involves asking for help, communicating difficulties, or proposing changes to the status quo (Choo, Linderman, and Schroeder 2007). If team members fear that they will be embarrassed or punished when exploring, they may avoid performing such activities (Choo, Linderman, and Schroeder 2007; Ooi et al. 2013). If, on the other hand, they feel they do not run interpersonal risks—in other words, have a strong sense of psychological safety—they are motivated to communicate different ideas (Lee, Swink, and Pandejpong 2011; Newman, Donohue, and Eva 2017) and more prone to ask for help (Love et al. 2019). Safe environments also encourage people to listen more attentively and provide more accurate feedback (Chung and Jackson 2013; Mom et al. 2015; Moran 2005; Ojha et al. 2018). Detailed explanations of ideas or problems can help individuals assimilate new information and achieve common comprehension, especially when the knowledge involved is complex (Akhavan and Mahdi, 2016; Yu, Zhang, and Huo

2020). Greater exposure to ideas (and problems) and an increased capacity to incorporate them increase the diversity of perspectives within a team, facilitating exploration (Chandrasekaran and Mishra 2012; Chung and Jackson, 2013; Maurer, Bartsch, and Ebers 2011). Hence, we propose the following:

Hypothesis 4: Psychological safety increases exploration in operational teams.

3.3. Standardization, team perception of support for innovation, and exploration

Due to the abstract nature of explorative (innovative) activities, organisations need to communicate that exploration is part of the workers’ daily routine and is a valuable activity. For this communication to be effective, however, workers need to be aware that both management systems (i.e. practices) and resources are committed to exploration (Amabile et al. 1996). For instance, Toyota’s *Kaizen* methodology guides continuous improvement by explicitly defining operational mistakes as a source of learning. *Kaizen* promotes practices that help teams identify, measure, and solve operational problems (Choo, Linderman, and Schroeder 2007). IDEO’s *Design Thinking* method guides the process of creating new solutions by promoting an empathic understanding of the problems that teams are trying to solve and providing processes to achieve the divergent and convergent thinking necessary for innovation (Wright, Sturdy, and Wylie 2012). Standardization is a means of communicating tacit concepts) and organizational values (Kirkhaug 2009). The shared perception of support for innovation enables team members to develop shared understandings of organizational priorities and goals towards innovation (Vlaar et al. 2006). We therefore hypothesize:

Hypothesis 5: Standardization is positively related to the shared perception of support for innovation in operational teams.

Standardization can lead to team exploration by affecting the social environment in which exploratory behaviours are performed. Standardization demands explicit rules and norms that prescribe exploratory behaviours as appropriate. Shared understandings of organizational culture and goals within teams facilitate individual actions and constrain undesirable behaviours, favouring collective interests (Coleman 1988). Simply put, if team members share that the organization supports innovation, the legitimacy and valuation of exploratory behaviours increase. Engaging in exploratory behaviours, however, can be hard for assembly line workers since these activities are uncertain and entail a high probability of failure (Khazanchi, Lewis, and Boyer 2007). If team members share that the organisation promotes innovation, these exploratory behaviours become legitimised within the team (Chen et al. 2007; Eiteneyer, Bendig, and Brettel 2019; Lechner, Frankenberger, and Floyd 2010; Higher legitimacy increases workers’ motivation because they perceive innovation as valuable and feasible (Hülshager, Anderson, and Salgado 2009). In turn, team members will be more available to work together in search of ways to improve processes. The

establishment of congruent understandings about culture and goals can guide the nature, direction, and magnitude of the parties' efforts. Furthermore, in work environments where there is a shared perception of support for innovation, unsuccessful initiatives are more likely to be tolerated (Hülshager, Anderson, and Salgado 2009), and tolerance of failure further motivates team members to engage in exploration (Kuratko, Hornsby, and Covin. 2014). Hence, we propose:

Hypothesis 6: The shared perception of support for innovation increases exploration in operational teams.

When we juxtapose the six hypotheses above, a mediating model emerges. Standardisation as a coordination mechanism indirectly affects team exploration by influencing the structural, relational, cognitive dimensions of social capital. Greater social capital means more opportunities to access and act upon new information and ideas. Teams using standardisation can improve exploration by increasing their social capital. Hence, we propose:

Hypothesis 7: The three social capital dimensions, (a) external communications, (b) psychological safety, and (c) support for innovation, mediate the positive relationship between standardization and exploration in operational teams.

4. Methods

4.1. Research settings and participants

Data was collected from 14 managers and 431 team members from 62 operational teams of a Chilean mining multinational. The company has a strong global presence in a wide variety of non-metallic products and generates approximately \$2 billion per year in revenues. The company implemented a Lean Production System in all of its plants. This Lean System strongly relies on the standardisation and rationalisation of activities (Yang, Lee, and Cheng 2017). Members of the various teams attended multifunctional meetings where they were strongly encouraged to participate in mapping, improving, and codifying organizational processes. Teams were geographically dispersed and worked in different operations of the company. The teams have discretion over how and what tasks are standardized. Upon completing this standardization, teams adhered to the codified processes and scheduled daily meetings to briefly assess operational outcomes and solve routine problems. Rules and templates for solving problems were made available as part of the Lean System implementation process. If team members identified complex problems, they could call extra meetings with people from outside their team.

The teams in this study worked twelve-hour shifts, rotating every seven days. The primary activities of these teams were to operate, control, and maintain various equipment and machines. Teams were quite independent; they were responsible for dividing their work, defining work strategies, and monitoring performance. Although operators worked alone, they were interdependent since each team member checked on others' work to control processes, performance, and safety standards. Compensation was based on team

performance. Finally, each team had a leader who reported to the Operations Manager of each plant.

We used two sources of information for this study: surveys administered to team members and their Lean System implementation process managers, and human resource data. The research project was presented to the teams during a general meeting, after which a questionnaire was distributed to each worker during their daily operational meetings. The research team informed workers of the academic focus of the study, discussed ethical issues, and insisted that participation was voluntary. Team members responded to the questionnaire without the presence of team leaders to minimise social desirability bias. Response rate was 80%.

The sample comprised 21 female (5%) and 410 male (95%) participants. With respect to the educational level, 60% of the participants possessed technical qualifications, and the remainder held bachelor's and master's degrees. Most workers had more than five years of experience at the company: 13% of participants had between 1 and 5 years of tenure, 31% between 5 and 10 years, 24% between 11 and 15 years, 19% between 16 and 20 years, and 13% more than 20 years. The mean team size was nine members, with a maximum of 16 and a minimum of four members. The average number of respondents per team was 6.9. We controlled for non-response bias using tenure and educational data. The t-test results showed no significant difference between respondents and non-respondents.

The project was also presented to the managers responsible for collectively providing support to the teams during the Lean implementation plan. Managers rotated between core teams and convened together daily for a progress meeting. During these meetings, each manager was asked to complete a short survey to rank the performance of 18 teams they had worked with. A total of 248 responses, with four responses per team, were received from these 14 managers.

4.1.2. Measures

As the research is concerned with team-level phenomena, we collected data at both the level of the team-member (level 1) and the team (level 2). All items and the reliability coefficients are detailed in [Appendix A](#).

Standardization: We measured standardisation using a scale from Gilson et al. (2005). This scale assessed the extent to which there were written procedures and guidelines for working situations. We adapted one question of the scale, asking specifically about written procedures and guidelines for explorative activities.

External team communication: We measured external team communication by asking how frequently team members communicated with members of other teams in the organisation. We utilised a three-item scale based on Ancona and Caldwell (1992). The frequency of interaction between intergroup members serves as a proxy of the strength of their relationship (Hill and Birkinshaw 2014).

Psychological safety: We used five items to measure psychological safety based on the scale developed by

Edmondson (1999), which assesses employees' perceptions of safety and respectability when they communicate ideas, problems, or concerns within their teams.

Shared perception of support for innovation: We measured team shared perception of innovation support using a five-item scale based on Kuratko's (2014) management support measure. The rationale for this measure was guided by extant literature. Specifically, Amabile et al. (1996) highlights that the perception of organisational support for innovation is aided by the encouragement of risk-taking and idea generation (*encouragement*), supportive and fair evaluation of new ideas (*idea generation*), and recognition of innovation or creativity (*recognition*). These aspects are reflected in the five-items utilized here.

Team exploration: To measure exploration, managers were asked to compare, and rate a given team against others, with a three-item scale from Patel (2011). A single team was rated by four different managers ($\alpha = 0.907$, ICC = 0.76).

Control variables: Relevant demographic variables such as team size, team tenure, the educational level of workers, and job function may influence a team's performance (Ancona and Caldwell 1992). Team size and job function were utilised as Level 2 control variables. We obtained the team size from team rosters, and we included the natural log of this number. This control is essential as larger groups may have more resources that can be allocated to exploration. Job function can also influence exploratory behaviours. Thus, we created dummy variables, one for the harvest process (HP), one for the chemical process (CP), and one for finished products (FP). We also control for individual educational level and tenure as Level 1 and 2 control variables. Higher education levels are positively associated with cognitive abilities to process information and learning, which may positively affect team members' exploration capacity (Mom et al. 2015). Team education was obtained by averaging the educational level of the team members (i.e. technician [1], bachelor's degree [2], and master's degree [3]). Team tenure refers to the levels of experience and is expected to negatively affect the team's exploration capacity (Gibson and Birkinshaw 2004). Team tenure was measured in terms of years of experience in ranges of seven years.

5. Results

5.1. Reliability and validity

Confirmatory factor analysis (CFA) was used to assess the data collected from team members. To test the data structure, we conducted a CFA with twenty-three items: standardisation (five items), team external communications (three items), psychological safety (five items), support for innovation (five items), and team exploration (three items). The resulting 5-factor and 21-item measurement model yielded acceptable goodness of fit ($\chi^2 = 309.07$; $df = 129$; $p < 0.00$; CFI = 0.951; TLI = 0.942; RMSEA = 0.061; SRMR = 0.048), with fit indices readings under suggested threshold values (Bentler and Bonett, 1980). Regarding reliability, Cronbach's alpha (α) was higher than 0.7, and composite reliability (CR)

ranged between 0.80 and 0.90, thus accomplishing the recommended value of 0.80 (Hair et al., 2014). Appendix A shows these results.

Convergent and discriminant validity were used to validate the variables of this study. Convergent validity was assessed by observing each item's coefficient on its underlying factor and with the average variance extracted. The standardized factor loading values of all construct items were above the recommended threshold 0.5, and their corresponding p -values were less than 0.01, indicating the items' significance (Bagozzi and Yi 1988). Moreover, the average variance extracted (AVE) values ranged between 0.57 and 0.65, above the recommended value of 0.50 (Fornell and Larcker, 1981).

Discriminant validity was examined through two approaches. First, a pair-wise test was performed for each pair of factors (Bagozzi and Yi 1988), testing whether CFA with two factors fits the data significantly better than a single-factor model. The scaled chi-squared difference values obtained for all pairs were statistically significant at the 5% level, thus demonstrating discriminant validity. Second, the values in parentheses in Table 1(a) indicate the squared root of AVE. These values are higher than the inter-correlation coefficient among constructs used in the model, indicating that discriminant validity has been achieved (Fornell and Larcker 1981).

Common-method variance. Self-report research designs risk capturing false correlations merely due to the use of a common source of data. To minimise the potential for common method variance, we employed different data sources and collected the independent and dependent variables at different times (c.f., Podsakoff et al., 2003). The independent variables and mediator are measured by team members, and the dependent variable is measured from surveys with managers. Two statistical analyses were later conducted to assess the severity of potential bias. The first was Harman's one-factor test. The factor analysis showed that a single general factor could explain 39.4% of the variance observed. The other method was the single-factor CFA. The results ($\chi^2 = 1701.83$; $df = 135$; $p < .00$; CFI = 0.573; TLI = 0.516; RMSEA = 0.1777; SRMR = 0.126; GFI = 0.455; AGFI = 0.450) suggest that the one-factor model provided very poor fit, suggesting that common method variance was not an issue.

5.2. Hypothesis tests

All hypotheses were tested simultaneously using a multilevel structural equation model (MSEM) in Mplus version 8 (Muthén and Muthén 1998–2017). The nature of the data is multilevel, as individuals are nested within teams. Failure to account for the interdependence in the data could lead to Type I and Type II error inflation in our estimations (Bliese 2002), which rules out the use of standard mediation approaches. As such, to calculate the indirect paths from standardisation to exploration through three mediators at the same time (α paths in the conceptual model, Figure 1), we employed MSEM (Preacher, Zyphur, and Zhang 2010), denoted as 1-(1,1,1)-2. In this model's notation, the first '1'

Table 1. Correlations Level 1 and Level 2.

Variables	M	SD	Level 1						
			1	2	3	4	5	6	7
1. Team Education	1.77	.76	1						
2. Team Tenure	2.85	1.23	-.26**	1					
3. Standardisation	4.52	1.25	.02	-.05	(0.77)				
4. P. Safety	3.87	.74	-.01	-.02	.36**	(0.77)			
5. S. Innovation	3.65	.91	-.07	-.04	.49**	.45**	(0.76)		
6. E.Communication	2.82	.97	.01	.10*	.38**	.36**	.43**	(0.78)	

Variables	M	SD	Level 2										
			1	2	3	4	5	6	7	8	9	10	11
1. Team Education	1.76	.45	1										
2. Team Tenure	2.87	.75	-.39**	1									
3. Team Size	2.05	.37	.03	.00	1								
4. Stock Piles (L1)	0.13	.34	.24	-.06	.08	1							
5. Chemical Process (L2)	0.52	.50	-.06	.06	-.02	-.40**	1						
6. Finished Products (L3)	0.35	.48	-.12	-.02	.15	-.29*	-.76**	1					
7. Standardisation	4.49	.92	.14	-.06	.06	.04	-.02	.01	1				
8. P. Safety	3.87	.45	.00	.01	.01	.01	-.17	.17	.49**	1			
9. S. Innovation	3.63	.58	-.13	.04	.08	.05	-.11	.08	.46**	.57**	1		
10. E.Communication	2.77	.54	.06	.16	.19	-.06	-.27*	.32**	.44**	.64**	.45**	1	
11. Exploration	3.99	1.01	-.08	.10	.09	.01	-.14	.13	.51**	.71**	.73**	.70**	1

* $p < 0.05$ ** $p < 0.01$.**Table 2.** Mediation model results from standardisation to team social capital.

Dependent Variable	H1 External Communication (a1 path)	H3 Psychological Safety (a2 path)	H5 Support for Innovation (a3 path)
Standardisation	0.262*** (0.078)	0.255*** (0.062)	0.253*** (0.060)
Observations	431	431	431

*** $p < 0.001$.

relates to the variable *standardisation*, measured at Level 1 (the team-member level). Next, '1,1,1' represents the three Level 1 mediating variables: *external communication*, *psychological safety*, and *support for innovation*. Finally, the '2' relates to the dependent variable, *team exploration*, measured at Level 2. Appendix A presents the Intraclass Correlation (ICC) values, providing support for the multilevel approach taken. The results that follow used maximum likelihood estimation and robust standard and were generated from a single 1-(1,1,1)-2 Mplus model.

Table 1 shows correlations at the within- and between-person levels.

Hypothesis 1, 3, and 5 predicted the relationship between standardisation and the mediating variables: external communication, psychological safety, and support for innovation (*a* paths) in the MSEM models. Table 2 displays these *a* paths at the between level of analysis. Standardization had a significant positive relationship with external communication (coeff. = .262, $p < .001$, 95% CI [.062, .463]), psychological safety (coeff. = .255, $p < .001$, 95% CI [.094, .416]), and support for innovation (coeff. = .253, $p < .001$, 95% CI [.098, .408]), supporting hypothesis 1, 3, and 5.

Hypothesis 2, 4, and 6 predicted a positive relationship between the mediator (external communication, psychological safety, and support for innovation) and team exploration (*b* paths). Model 2 in Table 3 revealed a positive and significant relationship between external communication and team exploration (coeff. = .1629, $p = .495$, 95% CI [.658, 2.60]), supporting hypothesis 2 (b1 path). Moreover, it showed a significant relationship between support for innovation and team exploration (coeff. = 1.457, $p < .001$, 95% CI

Table 3. Mediation model results towards team exploration.

	Model 1		Model 2	
	Estimate	SE	Estimate	SE
Direct effect				
Standardisation (c path)	.706***	(.136)	0.095	.156
External Communication (b1 path)			1.629***	.377
Psychological Safety (b2 path)			-.0875	.767
Support for Innovation (b3 path)			1.457***	.380
Indirect effect				
External Communication (ab1 path)			0.427*	0.165
Psychological Safety (ab2 path)			-.223	.214
Support for Innovation (ab3 path)			0.368**	0.134
Observations	431		431	
Log L	-1860.58		-3239.28	

This table presents c path, b paths (direct), and ab paths (indirect), as illustrated in Figure 1.

*** $p < 0.001$.

[.479, 2.436]), supporting hypothesis 6 (b3 path). Model 2, however, revealed a negative but not significant relationship between psychological safety and team exploration (coeff. = -.0875, $p = .254$, 95% CI [-.875, .254]), against what we predicted in Hypothesis 4 (b2 path).

Hypothesis 7 predicted that external communication, psychological safety, and support for innovation should mediate the relationship between standardisation and team exploration (combining *a* paths and *b* paths in Figure 1). Table 3 displays the indirect path estimates from standardization to team exploration through each mediator. The indirect path through external communication (Model 2 in Table 3) was positive and significant (coeff. = .427, $p < .05$, 95% CI [.002, .0853]), supporting Hypothesis 7a. The indirect path through psychological safety (Model 2 in Table 3) was negative but not significant (coeff. = -.223, $p = .297$, 95% CI [-.775,

.328]), rejecting Hypothesis 7b. The indirect path through support for innovation (Model 2 in Table 3) was positive and significant (coeff. = .368, $p < .01$, 95% CI [.024, .712]), supporting Hypothesis 7c.

5.3. Robustness check

To further examine the generalizability of the findings and the robustness of the unexpected findings concerning Hypothesis 7b, we conducted an additional study (Study 2) on a more diverse sample of workers. Participants were recruited online using Prolific. All 450 respondents work in teams (most of them between five and nine members and most working for more than two years together) and are based in different countries (primarily in North America and Europe). Most respondents have some tertiary education, and all speak English.

The same set of measures from Study 1 were used. Social capital attributes were operationalized with three proxy variables (external communication (EC), psychological safety (PS), and support for innovation (SI)). To determine if these variables are in line with other forms of operationalizing social capital, we examined the correlation between the chosen set of variables listed below and the variables used in extant research: structural dimension (adapted from Sun 2012 and Athuanne, 2007), relational dimension (adapted from Sun 2012), cognitive dimension (adapted from Wu 2008). These variables are described in Appendix A. We found strong positive correlations, providing us with some confidence in the adoption of these measures.

To confirm the direction and strength of the model, we examined the same hypothesis from Study One using PROCESS from SPSS, on this more diverse sample of workers. Support was found for hypothesis 1, 3, and 5, which relate to the relationship between standardisation and the mediating variables (EC: coeff. = .19, $p < .001$; PS: coeff. = .20, $p < .001$; and SI: coeff. = .18, $p < .001$). Support was also found for hypothesis 2, 4, and 6, which are concerned with the relationship between the mediating variables and exploration (EC: coeff. = .10, $p < .05$; PS: coeff. = .20, $p < .001$; and SI: coeff. = .48, $p < .001$).

Using the proposed mediators in a multiple mediator model with bootstrapped estimates (in Process SPSS), we examined the indirect path through PS and SI, which were positive and significant (PS: coeff. = 0.04 [0.015, 0.068]; SI: coeff. = 0.10 [0.032, 0.157]), supporting Hypothesis 7b, and 7c. The indirect path through EC was positive but non-significant. We also tested the hypothesised model using the alternative operationalization of social capital and found that the three variables mediated the relationship between standardisation and exploration.

The results are similar in our two studies, comprised of Spanish-speaking and English-speaking team members of different organisations and countries, thus strengthening our findings' generalizability.

5.4. Supplementary analyses

To understand the non-significant effect of psychological safety on exploration in Study 1, we examined the three-way interaction effect of psychological safety, support for innovation, and external communication on team exploration at level 2 (Appendix B shows the results). This analysis showed a positive and significant effect, which indicates that exploration levels are heightened when the relationships between all three dimensions of team social capital are higher. Thus, one possible explanation for the non-significant effect of psychological safety (PS) on team exploration is that PS must be combined with support for innovation and external communication to have an effect on exploration. External communications provide channels for the development of new ideas. The perception of support for innovation is an attribute that reflects a company's characteristics; thus, it may have a more significant effect on promoting explorative actions than intra-team dynamics. These results are aligned with the meta-analysis of team-level predictors of innovation at work developed by Hülshager, Anderson, and Salgado et al. (2009), who found that support for innovation and external communication are more powerful agents of innovative work behaviour within teams than psychological safety.

The three-way interaction results also suggest that each of the paths contribute synergistically to account for the relationship between standardisation and exploration. Standardisation increases the opportunity to access new ideas and the team's motivation to act on them. In other words, the absence of structural dimensions produces what Obstfeld (2005) called an *idea problem*: while workers might be motivated to explore, they might lack the connections to seek new ideas (Chung and Jackson 2013; Han 2018; Obstfeld 2005). Conversely, if workers are connected but do not sense psychological safety and support for innovation, they suffer from an *action problem*: they fail to act upon exploration opportunities (Obstfeld 2005; Tsai and Ghoshal, 1998).

6. Discussion and conclusions

This study developed a model that offers an alternative for solving the tensions between standardisation and exploration in operational teams. Based on social capital and duality theories, this study provides evidence to suggest that social capital mediates the relationship between standardisation and exploration. This work has theoretical and managerial implications. To theory, we extend our understanding of which social mechanisms positively influence team exploration under standardisation. In particular, we found that the cognitive and structural dimensions of social capital mediate the relationship between standardisation and exploration, which complements what has been previously described regarding the mediating role of psychological safety (i.e., the relational dimension) on exploration (e.g. Bressman and Zellmer-Bruhn 2012).

For practice, this study develops a logic for solving the problem of how to simultaneously motivate ideas and action

to facilitate team exploration when implementing process management programs (c.f., Burt 2007; Han 2018; Obstfeld 2005). We highlighted that the structural dimension of social capital might solve the idea problem by providing the opportunity to access new knowledge, while the cognitive and relational dimensions can solve the action problem by providing the motivation to act upon that new knowledge. Therefore, this research identifies new factors that mediate the relationship between standardisation and exploration and develops logic to explain how both ideas and actions can be simultaneously triggered in work teams.

6.1. Theoretical contributions

The debate of the effect between standardisation and exploration has evolved from a dualistic to a dual perspective. The dualistic perspective, which stems mainly from management literature, has posited a “either/or” view of the relationship between standardisation and exploration (Benner and Tushman 2003). Although this distinction facilitates understanding the phenomenon, the risk of framing the relationship as a dichotomy is ignoring more complex perspectives that can better capture the organisational reality (Farjoun 2010). In our case, we show that standardisation is not odd to exploration and can even improve it by enhancing the structural, relational, and cognitive dimensions of team social capital.

The operation management has framed standardisation as a dual mechanism that promotes both exploration and exploitation (Adler et al. 2009; Kristensen 2020). While the dual perspective allows capturing complex organizational phenomena, so far, the arguments for the effect of standardisation on exploitation have been cognitive or routine-based. Individual perspectives, however, fail to recognize that exploration resides on the coordination capabilities across individuals.

Building on the dual perspective, we extend the literature on operation management by showing that team social capital mediates the relationship between standardization and exploration. In doing so, we answer the call for more research exploring how process management programs affect innovative organisational capabilities (Evans, Foster, and Linderman 2014; Fundin et al. 2018; Shalley and Gilson 2017).

Furthermore, by focussing on teams, we contribute to the process management literature by offering a multilevel perspective on explorative and exploitative tensions. Most studies analysing the effects of PMPs' on exploration take either an individual or organisational perspective (Kim, Kumar and Kumar 2012; Kristensen 2020). Although research at the individual and organisational levels is fundamental to understand the organisational dynamics of exploration, this approach does not fully explain what drives differences in the behaviours and organisational explorative intentions (Argote and Miron-Spektor 2011).

The absence of a team meso-perspective can explain the discrepancies between dualism and duality. For example, dualism argues that standardisation constrains the

discretionary behaviours and cognitive abilities of individuals to explore (Benner and Tushman 2015). Duality studies, in contrast, have found a positive relationship between standardisation and firm innovative performance (Zeng et al. 2017). While in theory standardization can constrain individual behaviours, in practice we show that standardization promotes the external communication, psychological safety, and the perception of support for innovation in operational teams. Our multilevel (team) perspective thus contributes by disentangling how standardization affects team social dynamics, which can explain the positive effects of standardisation on firm innovative performance.

Finally, to the literature on social capital, we contribute by extending its use on operations management. Studies recently began using social capital to explain PMPs' effect on performance since organisational assets are embedded and mobilized through social relationships (Payne et al. 2011). These studies have examined the effect of social capital on six sigma project success (Easton and Rosenzweig 2015) and lean implementation success (Onofrei et al. 2019). We expand this emergent stream of literature by disentangling the effects of standardization on the structural, relational, and cognitive dimensions of social capital.

6.2. Managerial implications

Organizations implementing Process Management Programs should use standardization not only as a mechanism for identifying and applying exploitative best practices, but also for promoting exploration by improving the coordination and information flows between workers. Based on our results, managers should promote continual collaboration in the definition of those operational best practices and place more emphasis on sharing those standards. Doing so increases the social capital of working teams, the resources made available by their networks, and helps to balance explorative and exploitative demands.

The tools utilized for standardization should also be reviewed in light of our findings. Value stream mapping (VSM), for example, is a methodology to remove or reduce ‘waste’ in operational processes, thereby increasing efficiency and creating lean operations (Seth et al. 2017). An exclusive focus on efficiency, however, can hinder interactions between teams, which is crucial for transmitting tacit knowledge and the cross-fertilization of ideas (Roldán-Bravo et al. 2017). Therefore, standardization methodologies could also consider including networking and other activities to improve the social capital of working teams.

Standardization can help organizational change by facilitating the communication of new policies and visions towards improvement and innovation. This effect is particularly true for operators who are hierarchically distant from managers who define strategies. Managers implementing PMPs should recognize and take advantage of the opportunity to create shared understandings among workers, such as the perception of support for innovation.

Our findings also show that the workers' active involvement in the standardization process can improve their

psychological safety. Standardization can improve the sense of transparency, facilitating the perception of workers' safety (Kristensen 2020). Nevertheless, this psychological safety could be easily undermined if workers find the process has led to intensified work. Adler cites an example from Toyota. Taiichi Ohno – a key figure in the Toyota Production System development—was nicknamed Taiichi 'Oh no!' because, whenever he visited a plant, workers would stiffen in anticipation of yet another round of workforce reduction (Adler 2009). Managers should use standardization to provide workers with psychological safety and be receptive to situations when standardization outcomes can go against employee well-being.

6.3. Limitations and future research

Like all studies, this research is not without limitations. This study was restricted to members of operational teams who performed repetitive tasks. When tasks are variable, standardization could be useless or even counterproductive (Qamar et al., 2020; Zhang et al. 2012). For instance, the company where this study was conducted applied process management techniques to its research and development (R&D) activities with little success. R&D personnel complained that these processes limited their actions and creativity. Standardization as a set of rules may not be particularly useful as uncertainty increases, but standardisation in the form of guidelines could still be useful (Shalley and Gilson 2017). Moreover, standardization may be positive up until an inflection point and then produce non-linear effects, whereby high levels of standardization could thwart intentions to explore. Future research could determine which specific standardization practices and levels may promote productivity in uncertain environments.

We did not analyse how different operational environments moderate the relationship between standardisation and exploration concerning task design. For instance, some companies rely on high work-in-process inventories (WIPs) to make tasks less interdependent. In operational environments with high WIPs, the need for communication within the organisation is reduced and, therefore, the role of standardisation as a coordination mechanism may have a lesser effect (Adler et al. 2009). In contrast, low WIPs increase task interdependencies and demand more communication (Adler et al. 2009). For example, Toyota uses low inventory levels as a mechanism for exposing production problems and thereby fostering exploration (Osono et al. 2008). Future studies could examine how operational design or diverse use of technology (e.g., Elg et al. 2020; Roscoe and Blome 2019) that allows the use of low WIPs can moderate the effect of standardization on exploration.

Finally, given our cross-sectional design, we could not assess how the initial positive effect of standardization on social capital evolves or make specific causative claims. Over time, social networks can reinforce established behavioural expectations (Mom et al. 2015), create cognitive lock-ins (Lechner, Frankenberger, and Floyd 2010), increase redundant information sharing (Han 2018), or deepen existing

knowledge (Mom et al. 2015). Thus, longitudinal research and in-depth case studies will be required to understand the long-term consequences of standardization on firms, teams, and individuals' capacity to explore.

Disclosure statement

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Appendix A

Table A.1. Measurement items and data reliability index.

STUDY 1	
<i>Indicate the extent to which you agree with the following affirmations</i>	
<i>Response scale: 7-point scale, where 1 is for "strongly disagree" to 5 is for "strongly agree".</i>	
Standardisation (alpha = 0.90; CR = 0.90; AVE = 0.65; ICC = 0.437)	
STAN1	Team employed methods such as standard problem-solving procedures on a regular basis
STAN2	Team followed documented work sequences
STAN3	Performance appraisals are based on written performance standards
STAN4	There are established, formal communication channels
STAN5	Team had documented work processes
<i>Response scale: 5-point scale, where 1 is for "strongly disagree" to 5 is for "strongly agree".</i>	
Psychological Safety (alpha = 0.87; CR = 0.88; AVE = 0.59; ICC = 0.25)	
PSAFE1	I'm able to discuss problems and tough issues openly in this team
PSAFE2	If you make a mistake on this team, it is often held against you (reversed)
PSAFE3	It is difficult for me to ask other members of this team for help (reversed)
PSAFE4	No one on this team would deliberately act in a way to undermine my efforts
PSAFE5	Working with members of this team, my unique skills and talents are valued and utilised
Support for Innovation (alpha = 0.86; CR = 0.87; AVE = 0.58; ICC = 0.27)	
SINNOV1	In this team individuals with successful innovative ideas receive reward and recognition for their efforts
SINNOV2	People are encouraged to solve problems creatively in this team
SINNOV3	The team members who come up with innovative ideas on their own often receive management encouragement for their activities
SINNOV4	My team counts on mechanisms for developing new ideas
SINNOV5	Managers are receptive to my new ideas and suggestions
External Communication (alpha = 0.78; CR = 0.80; AVE = 0.57; ICC = 0.22)	
<i>Indicate the frequency with which you develop the following activities</i>	
<i>Response scale: 5-point scale, where 5 = daily, 4 = weekly, 3 = every two weeks, 2 = monthly, and 1 = rarely</i>	
ECOMM1	I share information with members from other teams within the organisation
ECOMM2	I am in touch with people of the organisation external to my team
ECOMM3	I engage in cross-functional activities
Team Exploration (alpha = 0.83; CR = 0.84; AVE = 0.637; ICC = 0.271)	
<i>Indicate the extent to which you agree with the following affirmations. During the past year ...</i>	
<i>Response scale: 1 (strongly disagree) to 5 (strongly agree).</i>	
TEXR1	Team members actively sought new operational possibilities to satisfy its customers' needs
TEXR2	Team members offered new ideas and solutions to complicated problems (were inventive)
TEXR3	Team members experimented with new and creative ways for accomplishing work
STUDY 2	
Relational dimension (alpha = 0.79; CR = 0.79) (adapted from Sum 2012)	
<i>Response scale: 5-point scale, where 5 = always, 4 = often, 3 = sometimes, 2 = seldom, and 1 = never</i>	
RELAT1	The relationship is characterised by personal friendship between employees in my team.
RELAT2	The relationship is characterised by mutual trust between employees in my team.
RELAT3	The relationship is characterised by high reciprocity between employees in my team.
RELAT4	Team members will not take advantage of others even when the opportunity arises.
Cognitive dimension (alpha = 0.77; CR = 0.78) (Wu 2008)	
COGN1	My colleagues in my team clearly understand the goals and vision in our company.
COGN2	My colleagues and I shared the same ambition.
COGN3	People in our team are enthusiastic about pursuing the collective goals and missions of the whole organisation.
Structural dimension (alpha = 0.87; CR = 0.87) (adapted from Sun, 2011; Atuhanne, 2007)	
<i>Response scale: 5-point scale, where 5 = always, 4 = often, 3 = sometimes, 2 = seldom, and 1 = never</i>	
STRUCT1	Employees in my team maintain close contact with other teams within the organisation.
STRUCT2	Employees of my team have social interaction with people of the organisation external to my team.
STRUCT3	Members of my team build relationships with other people in our organisation.
STRUCT4	Employees in my team spend a lot of time interacting with employees from other teams within the organisation.
STRUCT5	Employees in my team know some employees from other teams within the organisation.
STRUCT6	Employees in my team have frequent communication with employees of the organisation external to my team

Appendix B

Table B.1. Three-way interaction of psychological safety, support for innovation and external communication on team exploration.

	<i>Model</i>	
	<i>Coef.</i>	<i>Sig.</i>
Control		
Team Education	−0.066	.57
Team Tenure	−1.397	.25
Team Size	0.065	.78
Stock Piles	0.16	.51
Finished Products	0.12	.51
I.Variables		
Standardisation	0.02	.83
Mediators		
Psychological Safety (PS)	0.27	.29
Support for Innovation (SI)	0.73	.00
External Communication (EC)	0.48	.07
(PS)* (SI)	−0.04	.66
(PS)* (EC)	0.04	.67
(SI)* (EC)	0.09	.42
(PS)* (SI)* (EC)	0.17*	.04
R ²		0.70
F statistics		11.64